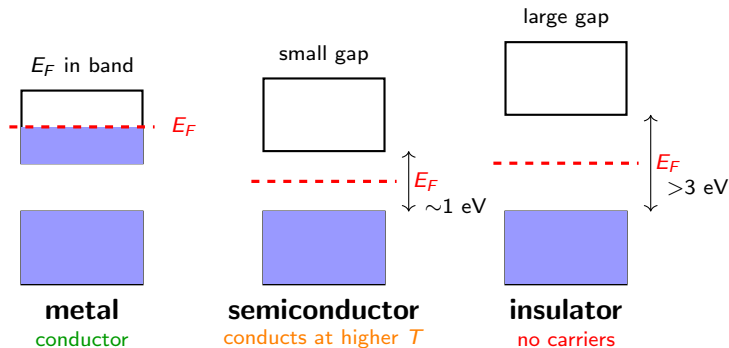


Lecture 13: Devices; P-N junction, diodes

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MAT SCI 120 — Winter 2026

Remember: materials and their band diagrams



General rule: electrical behavior is explained by mobility and band filling $\rightarrow$ band filling comes from the gap and E_F

Remember: Fermi-Dirac statistics and Boltzmann approximation

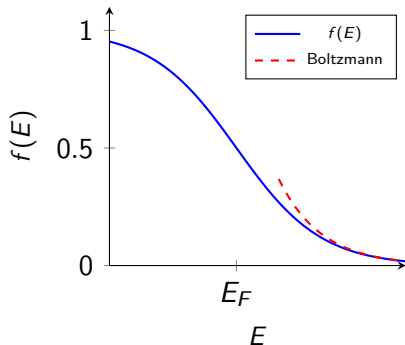
- Occupation probability:

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

- When E_F lies in gap far from edges:

$$f(E) \approx \exp\left(-\frac{E - E_F}{k_B T}\right)$$

- Valid for lower temperatures
($E - E_F \gg k_B T$)

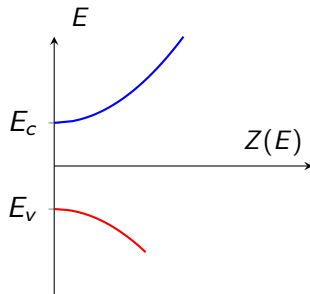


Remember: density of states in 3D parabolic bands

- ▶ Near band edges, $E(\mathbf{k})$ is parabolic
- ▶ DOS counts states per unit energy:

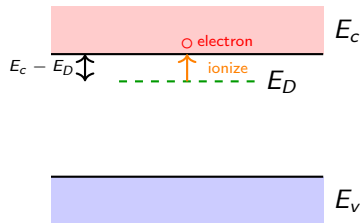
$$Z(E) = \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{3/2} \sqrt{E - E_c}$$

- ▶ Effective mass m^* accounts for band curvature



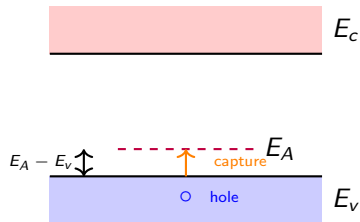
Remember: Donors add electrons to the conduction band (n-type)

- ▶ Group V atoms (P, As, Sb) in Si provide extra valence electron
- ▶ Extra electron loosely bound in hydrogen-like orbit
- ▶ Donor level E_D lies just below E_c (typically 10–50 meV)
- ▶ At room T : $k_B T \approx 26$ meV ionizes most donors



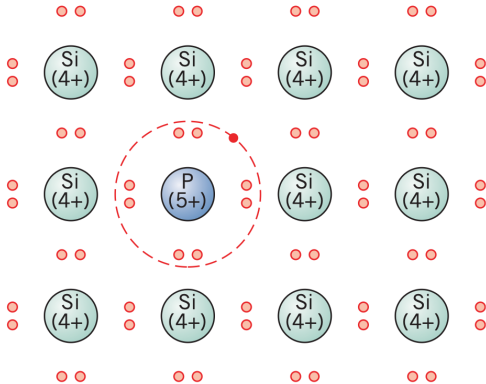
Remember: Acceptors add holes to the valence band (p-type)

- ▶ Group III atoms (B, Al, Ga) in Si: one fewer valence electron
- ▶ Acceptor captures electron from valence band $\rightarrow$ mobile hole
- ▶ Acceptor level E_A lies just above E_v (10–50 meV)
- ▶ Ionization: neutral atom $\rightarrow A^- + \text{hole}$

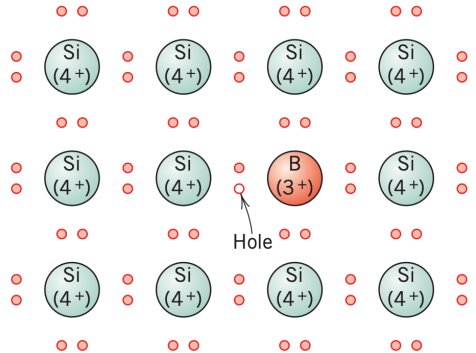


Remember: Donor vs acceptor picture

Donor (e.g., P in Si)

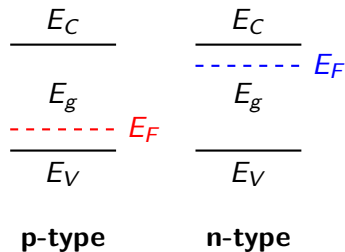


Acceptor (e.g., B in Si)



P-N junction: bringing two doped semiconductors together

- ▶ **p-type:** holes are majority carriers, Fermi level near valence band
- ▶ **n-type:** electrons are majority carriers, Fermi level near conduction band
- ▶ Contact creates interface where carriers must redistribute
- ▶ **Analogy with thermodynamics: Fermi level is a chemical potential!**
- ▶ **Analogy with kinetics: Fermi level is a concentration of electrons!**



Fermi level is a concentration of electrons: Fick's laws drive diffusion

- ▶ Carrier concentration set by Fermi level:

$$n = N_C \exp\left(\frac{E_F - E_C}{k_B T}\right)$$

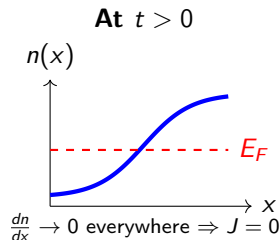
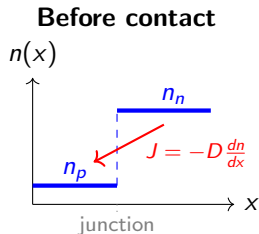
- ▶ Spatial variation in $E_F \Rightarrow \nabla n \neq 0$
- ▶ **Fick's 1st law** (flux $\propto$ gradient):

$$J = -D \frac{dn}{dx}$$

- ▶ **Fick's 2nd law** (evolution):

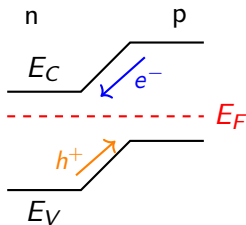
$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2}$$

- ▶ Equilibrium: $\frac{dn}{dx} = 0 \Leftrightarrow E_F = \text{const}$



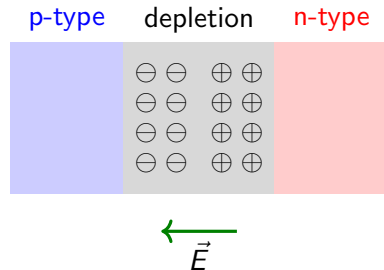
Fermi level equilibration at junction

- ▶ At equilibrium, Fermi level must be **constant** throughout the system
- ▶ Electrons diffuse from n to p; holes diffuse from p to n
- ▶ Diffusion leaves behind ionized dopants: **fixed charges** at interface



Depletion region formation

- ▶ Region near junction **depleted** of mobile carriers
- ▶ Fixed positive charges (N_D^+) on n-side, negative charges (N_A^-) on p-side
- ▶ Space charge creates internal electric field pointing from n to p
- ▶ Depletion region is charged because p-type and n-type semiconductors are **neutral**
- ▶ Why isn't the depletion region the entire semiconductor? → **electric field**
- ▶ How to find the electric field? → **Gauss' law**



Remember: Maxwell's equations (general)

With the definitions $\mathbf{D} = \varepsilon \mathbf{E}$ ($\mathbf{D}$ = displacement field) and $\mathbf{B} = \mu \mathbf{H}$ ($\mathbf{H}$ = magnetic field),

$$\nabla \cdot \mathbf{D} = \rho$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Poisson equation in depletion region

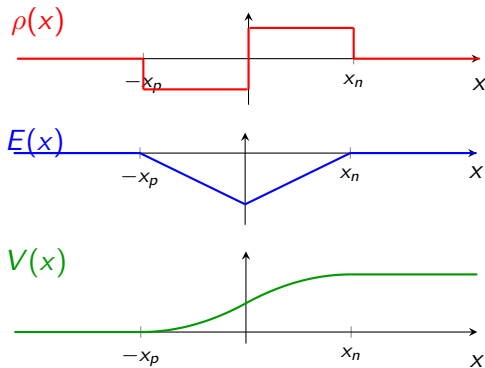
- Charge density in depletion approximation:

$$\rho(x) = \begin{cases} -eN_A & -x_p < x < 0 \\ +eN_D & 0 < x < x_n \end{cases}$$

- Poisson equation:

$$\frac{d^2 V}{dx^2} = -\frac{\rho}{\epsilon}$$

- Integrate once $\Rightarrow E(x)$ linear
- Integrate twice $\Rightarrow V(x)$ quadratic
- Boundary conditions: $E = 0$ and V continuous at $\pm x_{p,n}$



Derivation: Charge neutrality and electric field

Step 1: Charge neutrality defines depletion widths:

$$N_A x_p = N_D x_n$$

Step 2: Poisson equation $\frac{dE}{dx} = \frac{\rho}{\epsilon}$, integrate with $E(-x_p) = E(x_n) = 0$:

$$E(x) = \begin{cases} -\frac{eN_A}{\epsilon}(x + x_p) & -x_p < x < 0 \\ \frac{eN_D}{\epsilon}(x - x_n) & 0 < x < x_n \end{cases}$$

Continuity at $x = 0$ gives the peak field:

$$E_{\max} = \frac{eN_A x_p}{\epsilon} = \frac{eN_D x_n}{\epsilon}$$

Derivation: Electric potential and built-in voltage (V_{bi})

Step 3: Integrate $E = -dV/dx$ with $V(-x_p) = 0$:

$$V(x) = \begin{cases} \frac{eN_A}{2\epsilon}(x + x_p)^2 & -x_p < x < 0 \\ V_{bi} - \frac{eN_D}{2\epsilon}(x - x_n)^2 & 0 < x < x_n \end{cases}$$

Step 4: Evaluate at $x = 0$ to get the **built-in potential**:

$$V_{bi} = \frac{e}{2\epsilon}(N_A x_p^2 + N_D x_n^2)$$

Mass-action law and carrier concentrations

Mass-action law holds in each neutral region:

$$n p = n_i^2$$

Majority and minority carriers far from the junction:

$$\text{n-side: } n_n \approx N_D, \quad p_n = \frac{n_i^2}{N_D}$$

$$\text{p-side: } p_p \approx N_A, \quad n_p = \frac{n_i^2}{N_A}$$

Therefore the concentration ratio across the junction:

$$\frac{n_n}{n_p} = \frac{N_D}{n_i^2/N_A} = \frac{N_A N_D}{n_i^2}$$

Thermodynamics across the junction (part 1)

At equilibrium, E_F is constant everywhere. The electron concentration at any point:

$$n(x) = N_C \exp\left(\frac{E_F - E_C(x)}{k_B T}\right)$$

The conduction band edge shifts with the local electrostatic potential:

$$E_C(x) = E_{C,0} - eV(x)$$

Evaluate on each side of the junction:

$$n_n = N_C \exp\left(\frac{E_F - E_C(x_n)}{k_B T}\right), \quad n_p = N_C \exp\left(\frac{E_F - E_C(-x_p)}{k_B T}\right)$$

Thermodynamics across the junction (part 2)

Take the ratio (N_C and E_F cancel):

$$\frac{n_n}{n_p} = \exp\left(\frac{E_C(-x_p) - E_C(x_n)}{k_B T}\right) = \exp\left(\frac{eV_{bi}}{k_B T}\right)$$

What does this mean? A p-n junction created an imbalance on the concentration of electrons on the n side (n_n) and on the p side (n_p). This concentration is dependent on the built-in voltage and temperature.

But how to compute the built-in voltage?

Built-in voltage from thermodynamic argument

Combine the Boltzmann relation with the concentration ratio:

$$\frac{n_n}{n_p} = \frac{N_A N_D}{n_i^2} = \exp\left(\frac{eV_{bi}}{k_B T}\right)$$

Solve for V_{bi} :

$$V_{bi} = \frac{k_B T}{e} \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

This thermodynamic expression must equal the electrostatic result:

$$V_{bi} = \frac{e}{2\epsilon} (N_A x_p^2 + N_D x_n^2)$$

Equating the two gives a relation between the depletion widths x_p , x_n and the doping concentrations N_A , N_D .

Conceptual check: P-N junction basics

1. A Si p-n junction has $N_A = 10^{17} \text{ cm}^{-3}$ and $N_D = 10^{16} \text{ cm}^{-3}$. On which side is the depletion region wider? By what factor?
2. At equilibrium, is there a current flowing through the junction? Is there a diffusion flux? Is there a drift flux? Explain.
3. Compute V_{bi} for the junction above at $T = 300 \text{ K}$. Use $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$.
4. If the temperature increases, does V_{bi} increase or decrease? Consider what happens to $n_i(T)$.

Conceptual problems: non-trivial to try at home

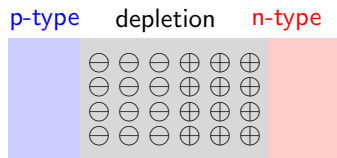
1. Starting from charge neutrality ($N_A x_p = N_D x_n$) and the electrostatic expression for V_{bi} , show that the total depletion width is:

$$W = x_p + x_n = \sqrt{\frac{2\epsilon V_{bi}}{e} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)}$$

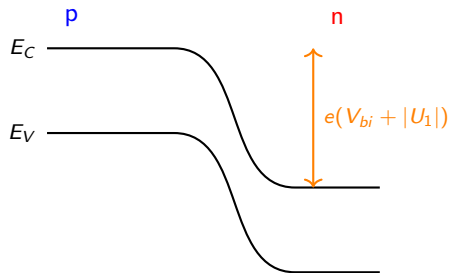
2. A p-n junction has $N_A = N_D = 10^{16} \text{ cm}^{-3}$. A second junction has $N_A = 10^{18}$ and $N_D = 10^{14} \text{ cm}^{-3}$. Compare V_{bi} for both. Which has a wider depletion region? In the asymmetric case, which side hosts nearly all of it?
3. The peak electric field is $E_{\max} = eN_A x_p / \epsilon$. For the asymmetric junction above, estimate $E_{\max}$ and W . Use $\epsilon = 12 \epsilon_0$.

Reverse bias: barrier increases, depletion widens

- ▶ Applied voltage $U_1 < 0$ adds to built-in potential
- ▶ Barrier becomes $e(V_{bi} + |U_1|)$
- ▶ Depletion region **widens**: fewer carriers cross the barrier
- ▶ The internal electric field $\vec{E}_{int}$ becomes stronger

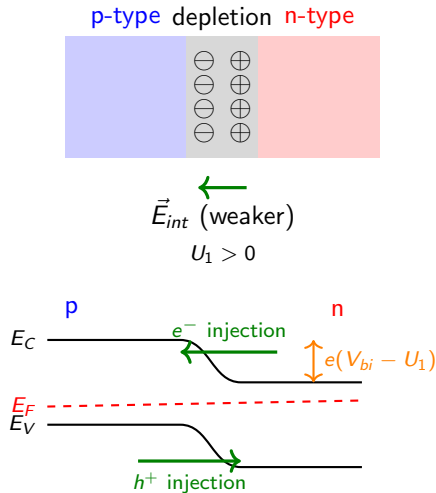


$\vec{E}_{int}$ (stronger)
 $U_1 < 0$



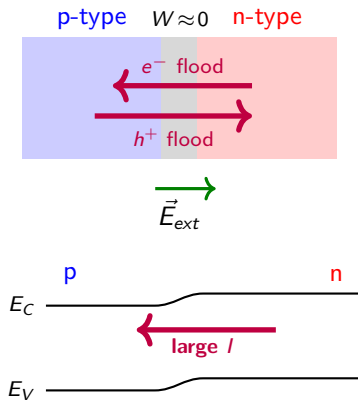
Forward bias: barrier decreases, depletion narrows

- ▶ Applied voltage $U_1 > 0$ opposes built-in potential
- ▶ Barrier reduced to $e(V_{bi} - U_1)$
- ▶ Depletion region **narrows**: majority carriers injected
- ▶ The internal electric field $\vec{E}_{int}$ becomes weaker
- ▶ Large diffusion current increases exponentially with U_1



Strong forward bias: ohmic conduction regime

- ▶ Applied voltage $U_1 \rightarrow V_{bi}$: barrier nearly eliminated
- ▶ Depletion region **collapses**: $W \rightarrow 0$
- ▶ Massive injection of majority carriers across junction
- ▶ The internal electric field $\vec{E}_{int}$ is negligible
- ▶ The external electric field $\vec{E}_{ext}$ drives conduction of electricity



Summary of P-N junctions

